

Lab Notebook Entry: HSF Evidence Script

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Script

hsf_substantiation_suite_v1.py

Purpose

This script was developed to replace a presentation-style HSF demonstration with explicit computational witnesses. The aim was to separate different claims and test them one by one, rather than rely on a single narrative script that bundled multiple ideas together.

The script is therefore best understood as an *evidence script*: it does not claim to derive all of HSF, but it does provide direct numerical checks for several core mechanisms.

What the Script Tests

1. Minimal Bidirectional Link Threshold

The script checks when a left-inherited action on a link admits an independent right copy in the commutant. The observed threshold occurs exactly at:

- SU(2): $d_B = 4$
- SU(3): $d_B = 9$
- SU(4): $d_B = 16$

This supports the clean threshold claim

$$d_B \geq N^2$$

for a minimal bidirectional link carrying independent inherited left/right structure.

2. Exact Single-Edge Gauge Invariance

A composite SU(3) single-edge attachment model is checked directly by computing commutators with local Gauss generators. The reported commutator norm is of order 10^{-15} , i.e. numerical zero.

This supports exact local attachment/gauge compatibility in the single-edge composite-link model.

3. Minimal Closed-Lattice Singlet Contrast

The script computes the trivalent local singlet dimension

$$\dim \text{Inv}(V \otimes V \otimes V)$$

and finds:

- SU(2): 0
- SU(3): 1

This gives a compact algebraic witness for why the SU(3) case admits a nontrivial local singlet while the analogous SU(2) case does not.

4. No-Refolding Witness

A random internal refolding is applied to a composite link in two ways:

1. **Incoherent refolding:** attachment generators are not updated.
2. **Coherent propagated refolding:** attachment generators are updated together with the refolding.

The result is sharp:

- incoherent refolding strongly breaks compatibility,
- coherent propagation preserves compatibility to numerical precision.

This provides a concrete witness for the no-refolding idea: once attachments exist, arbitrary internal rearrangements are not free.

5. Explicit HSF Link-Chain Witness

The script implements the chain

$$\text{Sub A} \leftrightarrow L_\ell \leftrightarrow T \leftrightarrow L_r \leftrightarrow \text{Sub B}$$

with

$$T = T_L \otimes T_R \otimes K,$$

where:

- L_ℓ inherits structure from subsystem A,
- L_r inherits structure from subsystem B,
- T_L and T_R are the active transmission channels,
- K is gauge-trivial slack.

5a. Bondwise Inherited-Structure Compatibility

Each adjacent bond is tested against the corrected diagonal inherited action on that bond:

- $A \leftrightarrow L_\ell$
- $L_\ell \leftrightarrow T_L$
- $T_L \leftrightarrow T_R$
- $T_R \leftrightarrow L_r$
- $L_r \leftrightarrow B$

All five reported commutator norms are of order 10^{-14} , i.e. numerical zero. This indicates that each adjacent bond is locally compatible with the inherited structure assigned to that bond.

5b. Bandwidth / Transmission Witness

The A-to-B transmission map is nondegenerate:

- $\|M\|_F$ is clearly nonzero,
- $\text{rank}(M) = 4$ for all tested slack sizes,
- $\|M(k) - M(1)\|_F$ remains tiny as slack grows.

Interpretation: enlarging uncoupled slack K increases Hilbert-space dimension but does *not* increase measured transmission rank. This is an explicit finite-bandwidth-style witness in the HSF link-chain architecture.

Observed Output Summary

The current run supports the following numerical conclusions:

1. The minimal bidirectional link threshold flips exactly at $d_B = N^2$.
2. The composite SU(3) single-edge gauge test passes to numerical precision.
3. The SU(2)/SU(3) trivalent singlet contrast is cleanly reproduced.
4. The no-refolding witness distinguishes sharply between incoherent and coherently propagated internal rearrangements.
5. The explicit HSF link chain produces a nonzero A-to-B transmission map whose rank remains fixed as slack grows.
6. The local adjacent inherited-structure checks pass bond by bond.

What This Script Now Supports

In limited witness form, the script supports:

1. Minimal bidirectional links require at least $d_B = N^2$ to carry clean independent left/right inherited structure.
2. A composite $SU(3)$ edge can satisfy exact local attachment/gauge compatibility.
3. $SU(3)$ admits a minimal trivalent singlet witness that $SU(2)$ does not.
4. No-refolding can be realized as a concrete structural restriction rather than only a verbal principle.
5. An explicit $A-L_\ell-T-L_r-B$ chain can carry nonzero transmission while remaining insensitive to added uncoupled slack.
6. Adjacent inherited structures in that chain can be locally compatible bond by bond.

What This Script Does *Not* Yet Show

This script does *not* yet establish:

- emergent factorization from bare Hilbert space,
- a full global node/gauge law over the entire transmitting chain,
- a final operational theorem for finite bandwidth,
- confinement observables such as Wilson loops, string tension, or area law.

Bottom Line

This script should be treated as an *evidence script*, not as a full derivation of HSF.

Its value is that it replaces broad narrative claims with explicit computational witnesses for several core ideas:

- inherited link structure,
- local compatibility,
- no-refolding,
- nontrivial transmission,
- slack-insensitive bandwidth behavior.

At present, this is a solid result package for an HSF lab notebook or evidence folder: it does not finish the framework, but it gives a defensible computational base for several of its central mechanisms.